

Case – Copper and China Industrial Pulse

Question + Mechanism – Does the copper price tend to move ahead on China Industrial Circle? Copper is often called “Dr. Copper” for its ability to predict economics turning points. We examine whether upturns or downturns in copper prices foreshadow similar move in composite industrial momentum index for China. The expected mechanism centers on the demand and inventory cycle. When manufacturers anticipate rising orders, they increase copper procurement(and restock inventories) in advance, lifting prices. Therefore, copper might rise before official indicator such as PMIs, output growth strengthen.

Data & Series – We construct an “Industrial Pulse” index for china by averaging a few key monthly indicators: 1) the official manufacturing PMI output sub index(AKA Production) 2) new orders(Especially export orders) from the PMI survey 3) YOY growth rate in merchandise export and 4) a freight activity proxy (ex. Railway, cargo volume index). This pulse capture both domestic factory conditions(PMI) and external demand (exports). We take monthly LME copper price(USD/Metric ton) and consider the 3 months change in copper(percent). We align copper leads/lags by shifting the copper series forward by 3 months

Analysis Procedure: Importing or real data from FRED API. For Industrial pulse analysis each pulse was z scored over its full history, then the three z score were average month by month and skipping month where information are not complete(only one or two component available). For copper signal we find the change over 3 month using $PCOPPUSDM.pct_change(3) \times 100$ which is $FREDID.pct_change(3) \times 100$. Then we find the correlation and lead by testing if copper level predict the industry pulse later.

Below is the correlation output after we import and calculate all the data.

Lag	Pearson r	n	Interpretation
0	+0.3492	290	contemporaneous
1	+0.3595	289	copper leads pulse by 1 month
2	+0.3411	288	copper leads pulse by 2 months
3	+0.3186	287	copper leads pulse by 3 months

Here the correlation are highest when copper lead the pulse by 1 months but there is only 0.01 gap higher than contemporaneous. Which is too small to assume that copper is a true leading indicator. Which give the “Dr.Copper” hypothesis too weak.

